**Source Code:**

```
packageservlet;import java.io.IOException;
import javax.servlet.ServletException; import
javax.servlet.annotation.WebServlet; import
javax.servlet.http.HttpServlet; import
javax.servlet.http.HttpServletRequest; import
javax.servlet.http.HttpServletResponse;
import java.io.*;
@WebServlet("/WebApplication1") public
class MyServlet extends HttpServlet { private
static final long serialVersionUID = 1L;
public MyServlet() {
  super();
    }
   protected void doGet (HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
response.setContentType("text/html");        PrintWriter out = response.getWriter();
out.print("<html><head>My First Servlet</head><body>");
out.print("<center><h2>");
 out.print("Welcome to HTTP Servlets</h2></center>");        out.print("<p>doGet()
Method is executing....</p>");
out.print("</body></html>");
    }
}
```

**Output:**

My First Servlet
Welcome to HTTP Servlets
doGet() Method is executing....

## Source Code:

## 1) HTML Form (index.html):

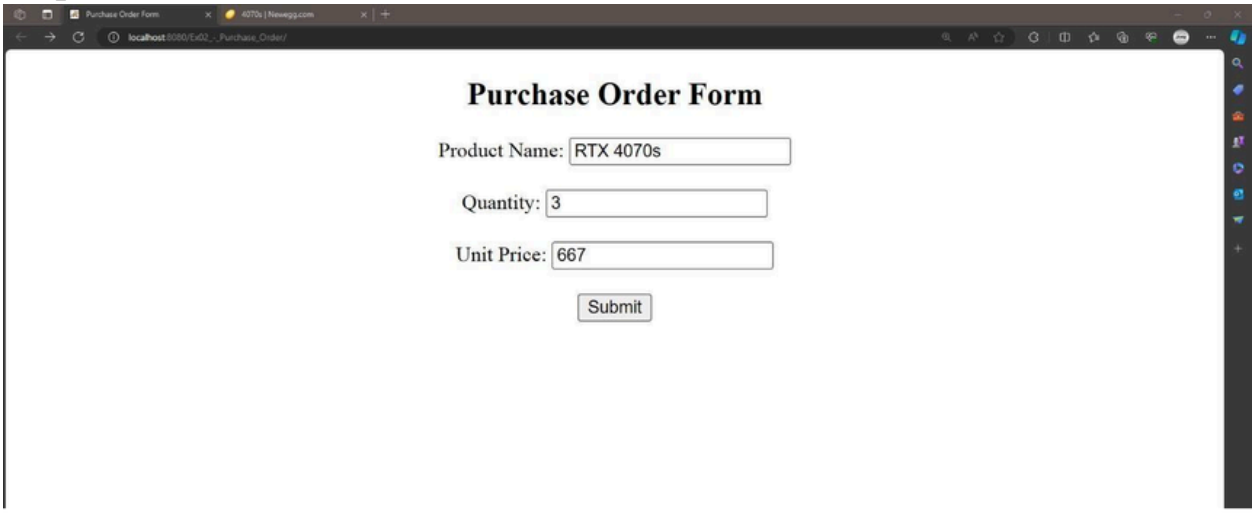
```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta http-equiv="X-UA-Compatible" content="IE=edge">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Purchase Order Form</title>
  </head>
  <body>
    <h2>Purchase Order Form</h2>
    <form action="PurchaseOrderServlet" method="post">
      <label for="productName">Product Name:</label>
      <input type="text" id="productName" name="productName" required><br><br>
      <label for="quantity">Quantity:</label>
      <input type="number" id="quantity" name="quantity" required><br><br>
      <label for="unitPrice">Unit Price:</label>
      <input type="number" id="unitPrice" name="unitPrice" step="0.01" required><br><br>
      <input type="submit" value="Submit">
    </form>
  </body>
</html>
```

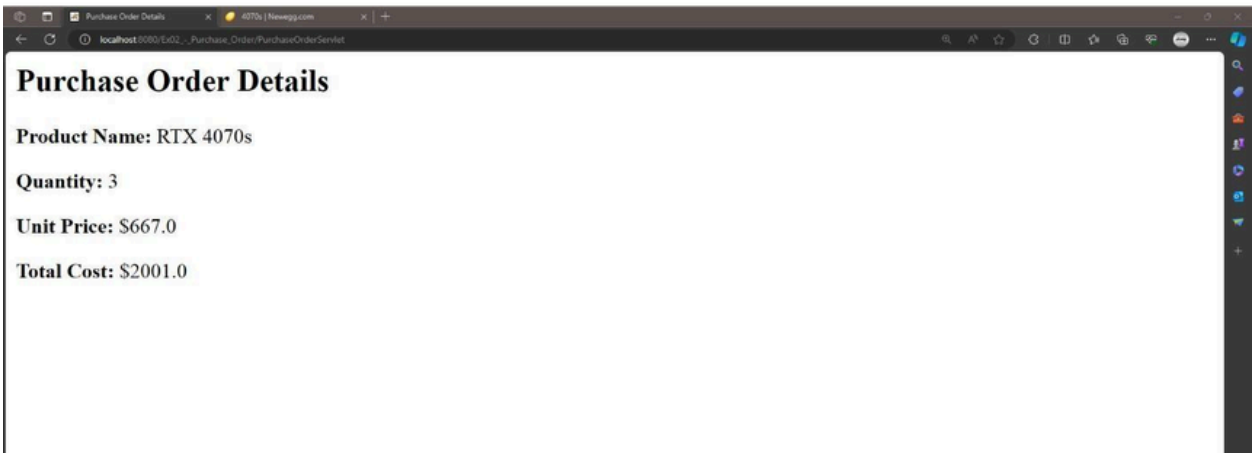
## 2) JAVA-Servlet(PurchaseOrderServlet.java):

```
importjava.io.IOException;import java.io.PrintWriter;import
javax.servlet.annotation.WebServlet; import javax.servlet.http.HttpServlet; import
javax.servlet.http.HttpServletRequest; import
javax.servlet.http.HttpServletResponse; @WebServlet("/PurchaseOrderServlet") public
class PurchaseOrderServlet extends HttpServlet {     private static final long
serialVersionUID = 1L;     protected void doPost(HttpServletRequest request,
HttpServletResponse response)            throws IOException {
    // Retrieve form data
```

```
        String productName = request.getParameter("productName");        int
quantity = Integer.parseInt(request.getParameter("quantity"));        double unitPrice
= Double.parseDouble(request.getParameter("unitPrice"));
        // Calculate total cost        double totalCost = quantity * unitPrice;        // Display purchase
order details        response.setContentType("text/html");        PrintWriter out
= response.getWriter();        out.println("<html><head><title>Purchase Order
Details</title></head><body>");        out.println("<h2>Purchase Order Details</h2>");
out.println("<p><strong>Product Name:</strong> " + productName + "</p>");
out.println("<p><strong>Quantity:</strong> " + quantity + "</p>"); out.println("
<p><strong>Unit Price:</strong> $" + unitPrice + "</p>"); out.println("<p>
<strong>Total Cost:</strong> $" + totalCost + "</p>"); out.println("</body>
</html>");
    }
}
```

**Output:**


Purchase Order Form
Product Name: RTX 4070s
Quantity: 3
Unit Price: 667
Submit


Purchase Order Details
Product Name: RTX 4070s
Quantity: 3
Unit Price: $667.0
Total Cost: $2001.0

**Source Code:**

## 1) HTML Form (index.html):

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Marks Percentage Calculator</title>
  </head>
  <body>
    <h2>Marks Percentage Calculator</h2>
    <form action="calculatePercentage.jsp" method="post">
      <label for="tamil">Tamil:</label>
      <input type="number" id="tamil" name="tamil" required><br><br>
      <label for="english">English:</label>
      <input type="number" id="english" name="english" required><br><br>
      <label for="maths">Maths:</label>
      <input type="number" id="maths" name="maths" required><br><br>
      <label for="science">Science:</label>
      <input type="number" id="science" name="science" required><br><br>
      <label for="social_science">Social Science:</label>
      <input type="number" id="social_science" name="social_science" required><br><br>
      <input type="submit" value="Calculate Percentage">

    </form>
  </body>
</html>
```

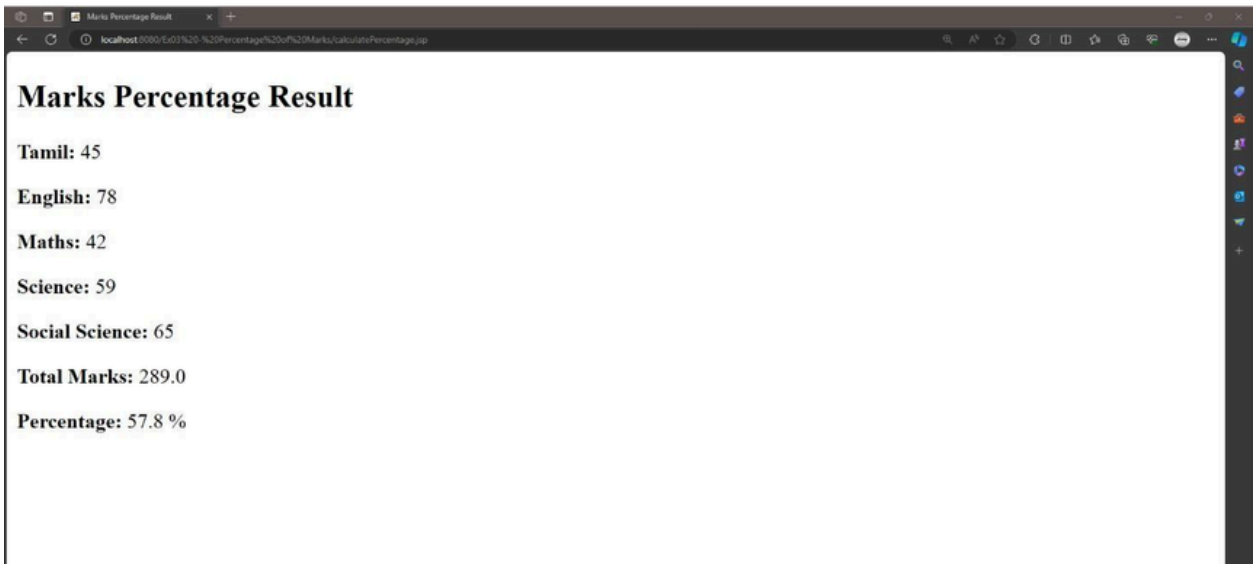
## 2) JSPFile (calculatePercentage.jsp):

```
<%@pagecontentType="text/html;charset=UTF-8" language="java" %> <% int
tamil=Integer.parseInt(request.getParameter("tamil"));      int english =
Integer.parseInt(request.getParameter("english"));      int maths =
Integer.parseInt(request.getParameter("maths"));     int science =
Integer.parseInt(request.getParameter("science"));     int social_science =
Integer.parseInt(request.getParameter("social_science"));     double totalMarks
```

```
= tamil + english + maths + science + social_science;     double percentage =
(totalMarks / 500) * 100;%>
<!DOCTYPE html>
   <head>
      <title>Marks Percentage Result</title>
   </head>
   <body>
      <h2>Marks Percentage Result</h2>
      <p> <strong>Tamil:</strong> <%= tamil%></p>
      <p> <strong>English:</strong> <%= english%></p>
      <p> <strong>Maths:</strong> <%= maths%></p>
      <p> <strong>Science:</strong> <%= science%></p>
      <p> <strong>Social Science:</strong> <%= social_science%></p>
      <p> <strong>Total Marks:</strong> <%= totalMarks%></p>
      <p> <strong>Percentage:</strong> <%= percentage%> % </p>
   </body>
</html>
```

## Output:

Marks Percentage Calculator
Tamil: 45
English: 78
Maths: 42
Science: 59
Social Science: 65
Calculate Percentage


Marks Percentage Result
Tamil: 45
English: 78
Maths: 42
Science: 59
Social Science: 65
Total Marks: 289.0
Percentage: 57.8 %

**Source Code:**

**1) HTML File (index.html):**

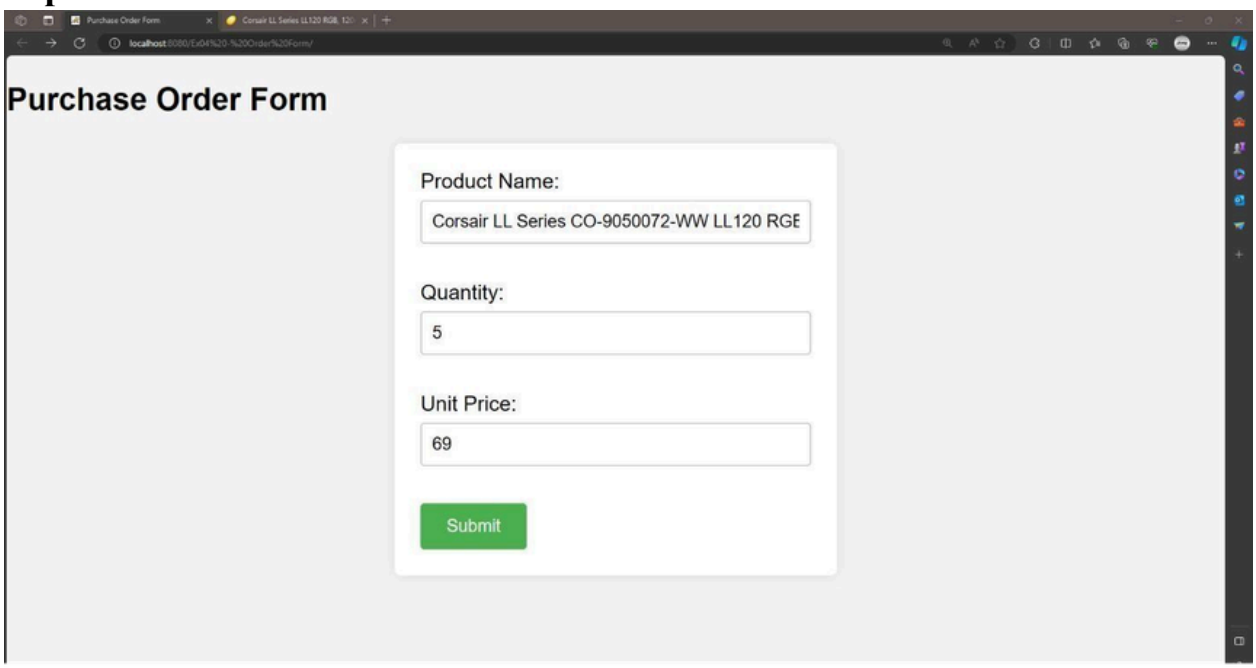
```
<!DOCTYPE html>
  <head>
    <title>Purchase Order Form</title>
  </head>
  <body>
    <h2>Purchase Order Form</h2>
    <form action="processPurchaseOrder.jsp" method="post">
      <label for="productName">Product Name:</label>
      <input type="text" id="productName" name="productName" required><br><br>
      <label for="quantity">Quantity:</label>
      <input type="number" id="quantity" name="quantity" required><br><br>
      <label for="unitPrice">Unit Price:</label>
      <input type="number" id="unitPrice" name="unitPrice" step="0.01" required><br><br>
      <input type="submit" value="Submit">
    </form>
  </body>
</html>
```

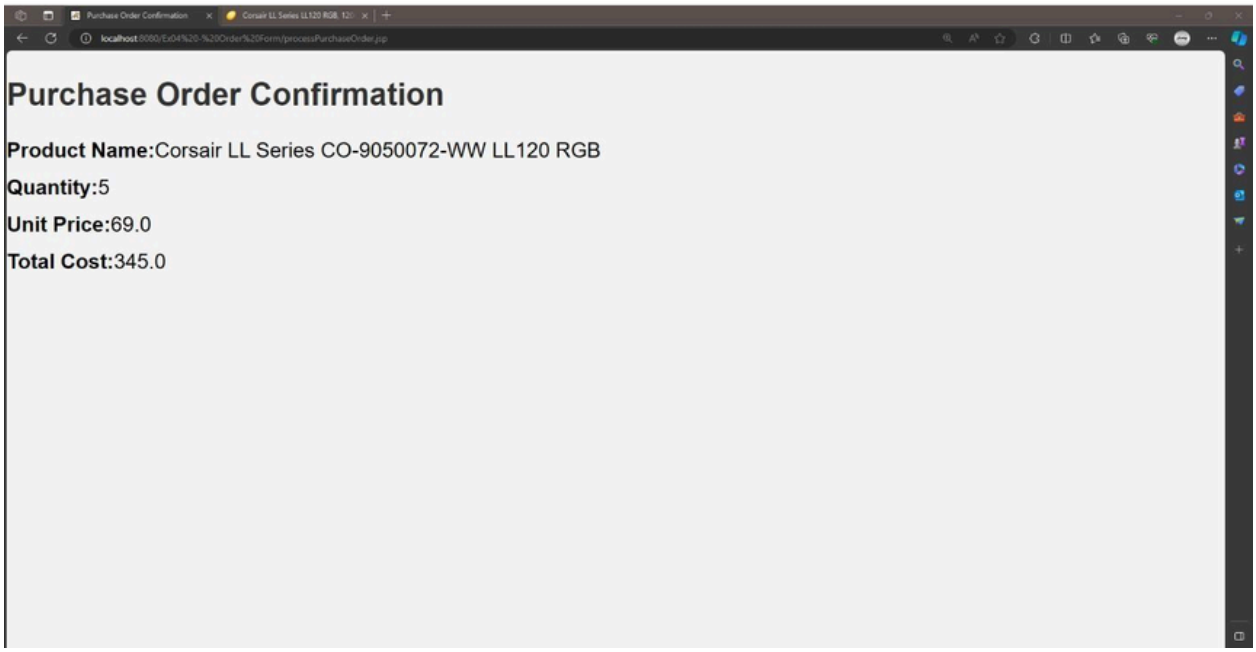
**2) JSPFile (processPurchaseOrder.jsp):**

```
<%@page contentType="text/html;charset=UTF-8"language="java" %>
<%@page import="java.util.*" %>
<% String productName = request.getParameter("productName");     int
quantity = Integer.parseInt(request.getParameter("quantity"));     double
unitPrice = Double.parseDouble(request.getParameter("unitPrice"));     double
totalCost = quantity * unitPrice;%>
<!DOCTYPE html>
  <head>
    <title>Purchase Order Confirmation</title>
  </head>
  <body>
    <h2>Purchase Order Confirmation</h2>
    <p><strong>Product Name:</strong><%= productName%></p>
```

```
 <p><strong>Quantity:</strong><%= quantity%></p>
        <p><strong>Unit Price:</strong><%= unitPrice%></p>
<p><strong>Total Cost:</strong><%= totalCost%></p>
    </body>
</html>
```

**Output:**


Purchase Order Form
Product Name:
Corsair LL Series CO-9050072-WW LL120 RGE
Quantity:
5
Unit Price:
69
Submit


Purchase Order Confirmation
Product Name:Corsair LL Series CO-9050072-WW LL120 RGB
Quantity:5
Unit Price:69.0
Total Cost:345.0

## Source Code:

## 1) HTML File (index.html):

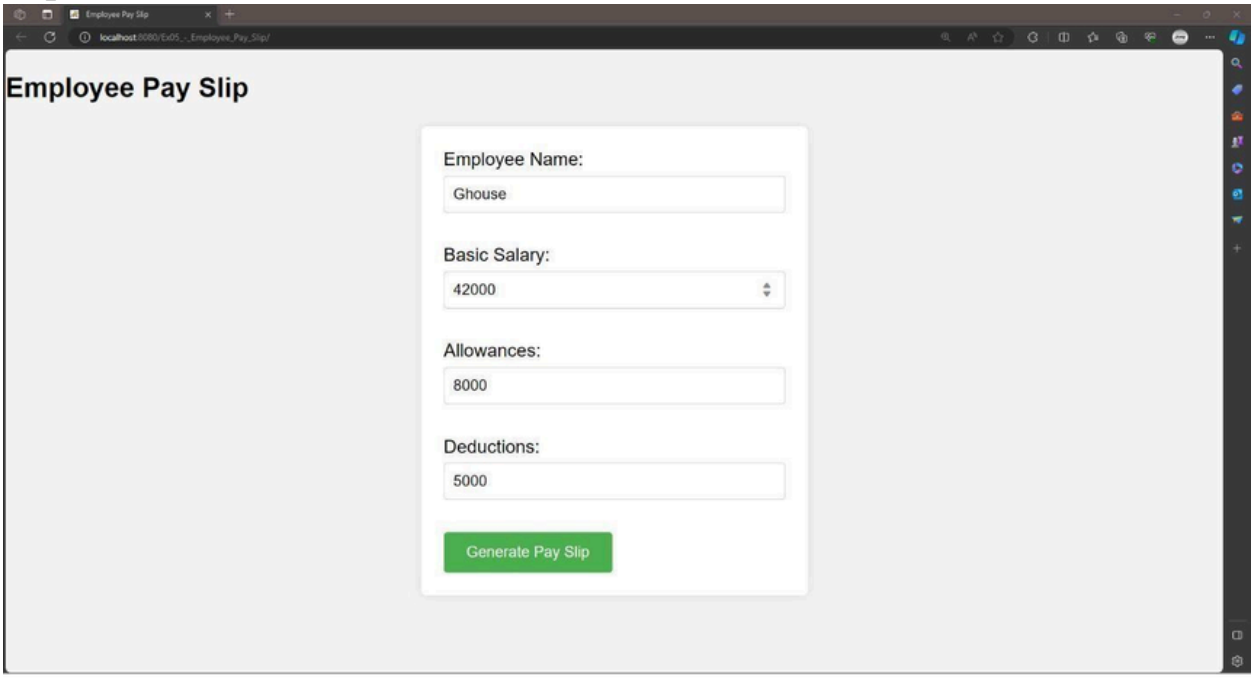
```
<!DOCTYPE html>
  <head>
    <title>Employee Pay Slip</title>
  </head>
  <body>
    <h2>Employee Pay Slip</h2>
    <form action="processPaySlip.jsp" method="post">
      <label for="employeeName">Employee Name:</label>
      <input type="text" id="employeeName" name="employeeName" required><br><br>
      <label for="basicSalary">Basic Salary:</label>
      <input type="number" id="basicSalary" name="basicSalary" required><br><br>
      <label for="allowances">Allowances:</label>
      <input type="number" id="allowances" name="allowances" required><br><br>
      <label for="deductions">Deductions:</label>
      <input type="number" id="deductions" name="deductions" required><br><br>
      <input type="submit" value="Generate Pay Slip">
    </form>
  </body>
</html>
```

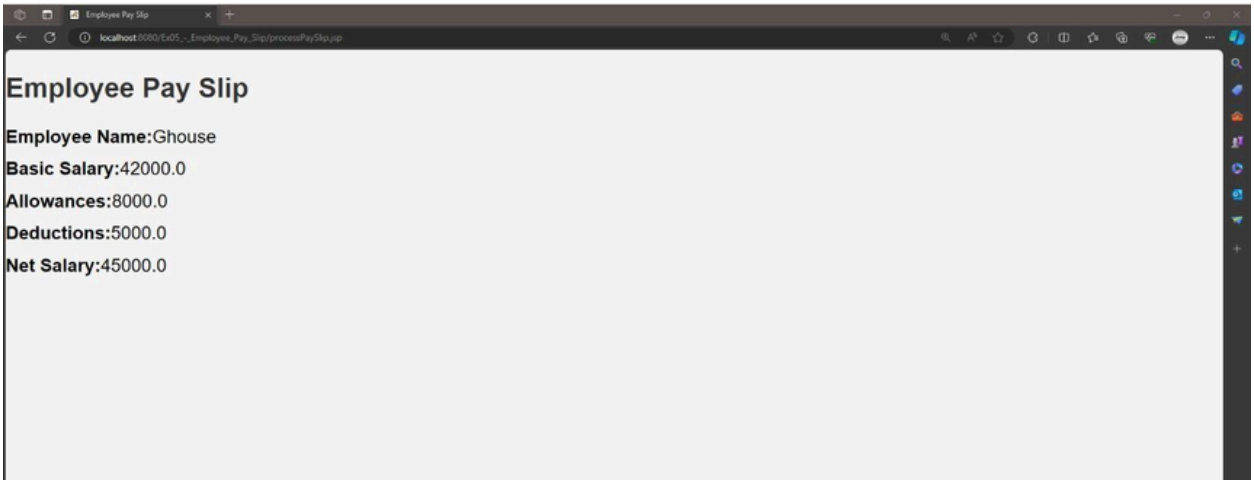
## 2)JSPfile (processPaySlip.jsp):

```
<%@pagecontentType="text/html;charset=UTF-8" language="java" %>
<%
  StringemployeeName = request.getParameter("employeeName");     double
basicSalary = Double.parseDouble(request.getParameter("basicSalary"));     double
allowances = Double.parseDouble(request.getParameter("allowances"));     double
deductions = Double.parseDouble(request.getParameter("deductions"));     double
netSalary = basicSalary + allowances - deductions;
%>
<!DOCTYPE html>
  <head>
    <title>Employee Pay Slip</title>
```

```
    </head>
    <body>
        <h2>Employee Pay Slip</h2>
        <p><strong>Employee Name:</strong><%= employeeName%></p>
        <p><strong>Basic Salary:</strong><%= basicSalary%></p>
        <p><strong>Allowances:</strong><%= allowances%></p>
        <p><strong>Deductions:</strong><%= deductions%></p>
        <p><strong>Net Salary:</strong><%= netSalary%></p>
    </body>
</html>
```

# Output:


Employee Pay Slip
Employee Name:
Ghouse
Basic Salary:
42000
Allowances:
8000
Deductions:
5000
Generate Pay Slip


Employee Pay Slip
Employee Name:Ghouse
Basic Salary:42000.0
Allowances:8000.0
Deductions:5000.0
Net Salary:45000.0

## Source Code:

## 1) Servlet (FormDataHandlerServlet.java):

```
importjava.io.IOException; import
java.io.PrintWriter; import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet; import
javax.servlet.http.HttpServlet; import
javax.servlet.http.HttpServletRequest; import
javax.servlet.http.HttpServletResponse;
@WebServlet(name = "FormDataHandlerServlet", urlPatterns = {"/FormDataHandlerServlet"})
public class FormDataHandlerServlet extends HttpServlet {     protected void
doPost(HttpServletRequest request, HttpServletResponse response)            throws
ServletException, IOException {         try (PrintWriter out = response.getWriter()) {
String name = request.getParameter("name");
            String email = request.getParameter("email");             String
message = request.getParameter("message");
out.println("<!DOCTYPE html>");
out.println("<html>");             out.println("<head>");
out.println("<title>Form Data</title>");
out.println("</head>");             out.println("<body>");
            out.println("<h1>Form Data</h1>");
out.println("<p>Name: " + name + "</p>");             out.println("<p>Email:
"+email+"</p>");                out.println("<p>Message: " + message +
"</p>");            out.println("</body>");            out.println("</html>");
        }
    }
}
```

## 2) HTML File (index.html):

```
<!DOCTYPEhtml>
<html>
    <head>
        <title>Form Submission</title>
    </head>
```

```
    <body>
        <h1>Submit Form</h1>
        <form action="FormDataHandlerServlet" method="post">
            <label for="name">Name:</label><br>
            <input type="text" id="name" name="name"><br>
            <label for="email">Email:</label><br>
            <input type="email" id="email" name="email"><br>
            <label for="message">Message:</label><br>
            <textarea id="message" name="message"></textarea><br><br>
            <input type="submit" value="Submit">
        </form>
    </body>
</html>
```

**Output:**

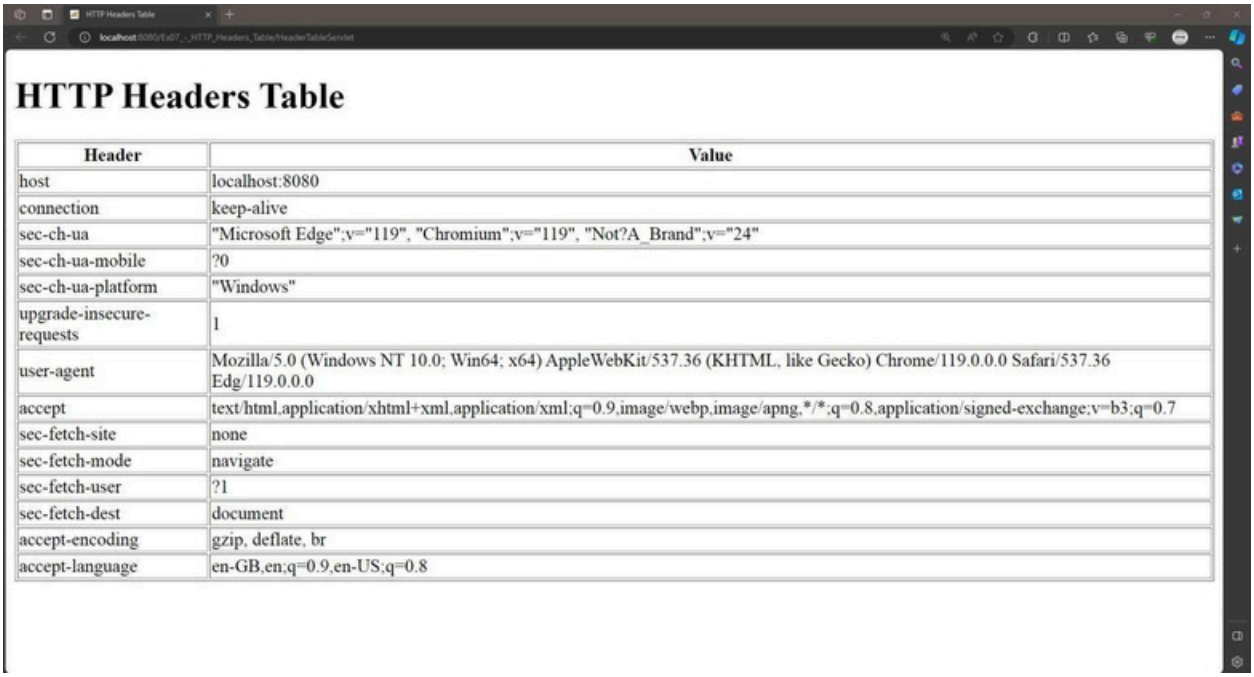
Submit Form
Name:
Ghouse
Email:
Message:
Submit

Form Data
Name: Ghouse
Email:
Message: It's enough Varsha, save some reels for tomorrow.

**Source Code:**

```
importjava.io.IOException; import
java.io.PrintWriter;import java.util.Collections;
import java.util.Enumeration; import
javax.servlet.ServletException; import
javax.servlet.annotation.WebServlet; import
javax.servlet.http.HttpServlet; import
javax.servlet.http.HttpServletRequest; import
javax.servlet.http.HttpServletResponse;
@WebServlet(name = "HeaderTableServlet", urlPatterns = {"/HeaderTableServlet"})
public class HeaderTableServlet extends HttpServlet {     protected void
doGet(HttpServletRequest request, HttpServletResponse response)            throws
ServletException, IOException {
response.setContentType("text/html;charset=UTF-8");        try (PrintWriter out =
response.getWriter()) {            out.println("<!DOCTYPE html>");            out.println("<html>");
out.println("<head>");
out.println("<title>HTTP Headers Table</title>");              out.println("</head>");
out.println("<body>");                  out.println("<h1>HTTP Headers Table</h1>");
out.println("<table border='1'>");
          out.println("<tr><th>Header</th><th>Value</th></tr>");
Enumeration<String> headerNames = request.getHeaderNames();            while
(headerNames.hasMoreElements()) {
              String headerName = headerNames.nextElement();
              Enumeration<String> headerValues = request.getHeaders(headerName);
while (headerValues.hasMoreElements()) {
                  String headerValue = headerValues.nextElement();
out.println("<tr><td>" + headerName + "</td><td>" + headerValue + "</td></tr>");                }
          }
          out.println("</table>");
out.println("</body>");            out.println("</html>");
      }
   }
}
```

## Output:



# HTTP Headers Table

| Header | Value |
|---|---|
| host | localhost:8080 |
| connection | keep-alive |
| sec-ch-ua | "Microsoft Edge";v="119", "Chromium";v="119", "Not?A_Brand";v="24" |
| sec-ch-ua-mobile | ?0 |
| sec-ch-ua-platform | "Windows" |
| upgrade-insecure-requests | 1 |
| user-agent | Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/119.0.0.0 Safari/537.36 Edg/119.0.0.0 |
| accept | text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7 |
| sec-fetch-site | none |
| sec-fetch-mode | navigate |
| sec-fetch-user | ?1 |
| sec-fetch-dest | document |
| accept-encoding | gzip, deflate, br |
| accept-language | en-GB,en;q=0.9,en-US;q=0.8 |

**Source Code:**

**1) Remote Interface: (CollegeInterface.java):**

```
importjava.rmi.Remote;import
java.rmi.RemoteException; public interface CollegeInterface
extends Remote {
   String getStudentInfo(int studentId) throws RemoteException;
}
```

**2) Server Class: (CollegeServer.java):** import

```
java.rmi.RemoteException; import java.rmi.registry.LocateRegistry; import
java.rmi.registry.Registry; import
java.rmi.server.UnicastRemoteObject; import java.util.HashMap; import
java.util.Map;
public class CollegeServer extends UnicastRemoteObject implements
CollegeInterface {     private Map<Integer, String> studentRecords;
public CollegeServer() throws RemoteException {         super();
       studentRecords = new HashMap<>();
studentRecords.put(5, "Eshwari");         studentRecords.put(8,
"Kalaiarasi");         studentRecords.put(20, "Savitha");
   }
   public String getStudentInfo(int studentId) throws RemoteException {
return studentRecords.getOrDefault(studentId, "Student not found");
   }
   public static void main(String[] args) {
try {
          CollegeServer obj = new CollegeServer();
          Registry registry = LocateRegistry.createRegistry(1099);
registry.rebind("CollegeServer", obj);
          System.out.println("Server ready");
       } catch (Exception e) {
          System.err.println("Server exception: " + e.toString());
e.printStackTrace();
       }
   }
}
```

### 3) Client Class: (CollegeClient.java)):

```
import java.rmi.registry.LocateRegistry; import
java.rmi.registry.Registry; import
java.util.Scanner; public class CollegeClient {
public static void main(String[] args) {        try
{
        Registry registry = LocateRegistry.getRegistry("localhost", 1099);
        CollegeInterface stub = (CollegeInterface) registry.lookup("CollegeServer");
        Scanner scanner = new Scanner(System.in);
System.out.print("Enter student ID: ");            int studentId
=scanner.nextInt();
        String studentInfo = stub.getStudentInfo(studentId);
        System.out.println("Student information: " + studentInfo);
      }catch (Exception e) {
        System.err.println("Client exception: " + e.toString());
e.printStackTrace();
      }
   }
}
```

# Output:

```
run:
Enter student ID: 5
Student information: Eshwari
BUILD SUCCESSFUL (total time: 2 minutes 17 seconds)
```

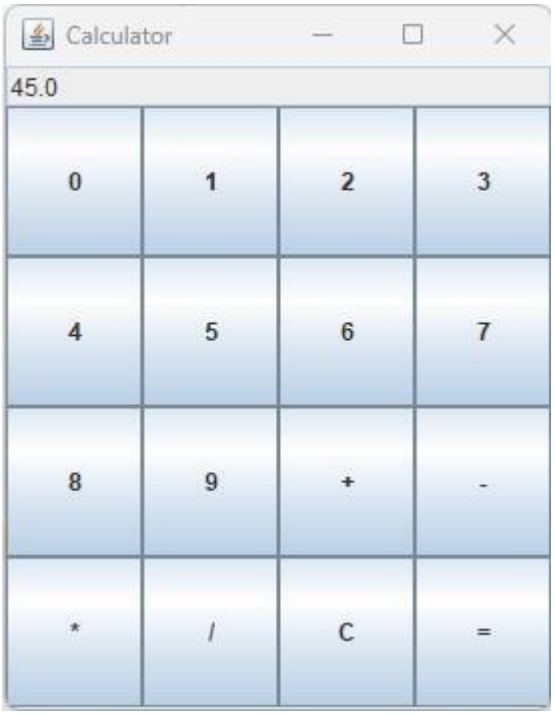
```
run:
Server ready
```

## Source Code:

```
importjava.awt.*;import java.awt.event.*; import javax.swing.*;
publicclassCalculator extends JApplet implements ActionListener {
privateJTextFieldtextField; private JButton[] numberButtons;
privateJButton[]operatorButtons; private JButton clearButton;
privateJButtonequalsButton; private String displayText = "";
privatedoubleresult= 0; private String operator = ""; public void
init() { setLayout(new BorderLayout()); textField = new
JTextField(); textField.setEditable(false); add(textField,
BorderLayout.NORTH); JPanel buttonPanel = new JPanel();
buttonPanel.setLayout(new GridLayout(4, 4)); numberButtons =
new JButton[10]; for (int i = 0; i < 10; i++) {
numberButtons[i] = new JButton(String.valueOf(i));
numberButtons[i].addActionListener(this);
buttonPanel.add(numberButtons[i]);
}
operatorButtons = new JButton[4];
operatorButtons[0] = new JButton("+");
operatorButtons[1] = new JButton("-"); operatorButtons[2]
= new JButton("*"); operatorButtons[3] = new
JButton("/"); for
(int i = 0; i < 4; i++) {
operatorButtons[i].addActionListener(this); buttonPanel.add(operatorButtons[i]);
}
clearButton = new JButton("C");
clearButton.addActionListener(this); buttonPanel.add(clearButton);
equalsButton
= new JButton("=");
equalsButton.addActionListener(this);
buttonPanel.add(equalsButton);
add(buttonPanel, BorderLayout.CENTER);
}
public void actionPerformed(ActionEvent e) {
JButton source = (JButton) e.getSource();
```

```
StringbuttonText= source.getText();          if
(buttonText.equals("C")) {            displayText=
"";            result =0;             operator ="";
       } else if (buttonText.equals("=")){            double
secondOperand= Double.parseDouble(displayText);            switch
(operator) {                case"+":                    result +=secondOperand;
break;                case"-":
                result -= secondOperand;
break;                case"*":
                result *= secondOperand;
break;                case"/":
                if (secondOperand != 0) {
result /=secondOperand;                    }else
{                    displayText= "Error:
Division by zero";
                }
break;
           }
           displayText = String.valueOf(result);
operator = "";
       } else if (buttonText.equals("+") || buttonText.equals("-")
|| buttonText.equals("*") ||buttonText.equals("/")) {            operator
=buttonText;            result =
Double.parseDouble(displayText);            displayText= "";
       }else {            displayText
+=buttonText;
       }
       textField.setText(displayText);
    }
}
```

**Output:**



**Source Code:**

**1) Receiver Project (Main.java):**

```
import
java.io.DataOutputStream; import
java.net.Socket; public class Main {     public
static void main(String[] args) {        try {
        String messageToSend = "Que Ota? (Translation: Want another one?)";
        Socket socket = new Socket("127.0.0.1", 12345);
        DataOutputStream out = new DataOutputStream(socket.getOutputStream());
outwrite(Massagetae);
        out. flush();
out.close();            socket.close();
        System.out.println("Message sent successfully.");
}catch(Exception e) {              e.printStackTrace();
      }
   }
}
```

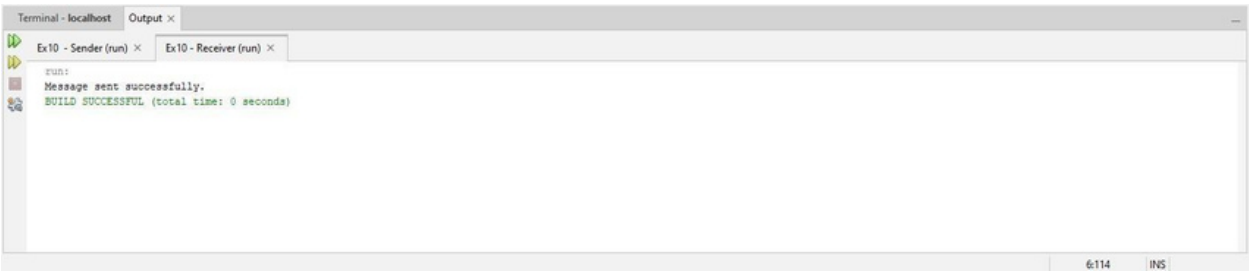
**2) Sender Project (Main.java):**

```
 import java.io.DataInputStream; import
java.net.ServerSocket;                import
     java.net.Socket; public class Main {
  public static void main(String[] args) {
                                        try {
        ServerSocket serverSocket = new ServerSocket(12345);
        System.out.println("Waiting for incoming connection...");
        Socket socket = serverSocket.accept();
        DataInputStream in = new DataInputStream(socket.getInputStream());
        String receivedMessage = in.readUTF();
        System.out.println("Received message: " + receivedMessage);
        }
in.close();            socket.close();            serverSocket.close();
catch(Exception e) {              e.printStackTrace();
      }
```

```
    }
}
```

# Output:

Terminal - localhost
Output - Ex10 - Sender (run)
run:
Waiting for incoming connection...
Ex10 - Sender (run)
17:33
INS


Terminal - localhost
Output
Ex10 - Sender (run)
Ex10 - Receiver (run)
run:
Message sent successfully.
BUILD SUCCESSFUL (total time: 0 seconds)
6:114
INS

Terminal - localhost
Output
Ex10 - Sender (run)
Ex10 - Receiver (run)
run:
Waiting for incoming connection...
Received message: Dr. Mann, do not open the inner hatch. I repeat, do not open the inner hatch!
BUILD SUCCESSFUL (total time: 2 minutes 13 seconds)